About this PDF

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Covers 15 of 17 syllabus sub-topics of this unit.

Not in the source (not covered in this PDF):

- HDLC
- Virtual LAN

Data Link Layer

Definition

The data link layer (layer 2) sits between the network layer above and the physical layer below and is often considered together with the physical layer, with Ethernet being the predominant protocol. It is divided into two sub-layers: the Medium Access Control (MAC) sub-layer, which handles media access, and the Logical Link Control (LLC) sub-layer, which defines the service access point to the upper layer.

Explanation

The data link layer receives the network-layer packet (including its header and trailer) as payload, determines the next-hop address, and forwards the data to the appropriate device. The MAC sub-layer interacts with the underlying physical medium, managing negotiation and framing, while the LLC sub-layer concerns itself with communication to higher layers and provides services such as unacknowledged connection-less delivery (best-effort). It must support a shared-medium, multi-access environment, allowing multiple stations to use the same link without requiring intermediate switching nodes.

Example

A LAN card (network adapter) implements the data link layer functions and part of the physical layer, enabling a PC to transmit an Ethernet frame that contains source and destination MAC addresses together with the payload from higher layers.

Key points

- The data link layer is split into MAC (media access) and LLC (service interface) sub-layers.
- It receives network-layer packets, adds its own header/footer, and forwards them to the next hop based on the destination address.
- It must support multi-access on a shared medium, allowing several stations to communicate over the same LAN.
- LAN adapters provide the practical implementation of these data link layer functions.

Framing

Key points

- The receiver uses a single variable **R** to hold the sequence number of the next expected frame; a frame is accepted only if its sequence number matches **R**, otherwise it is rejected.
- Positive acknowledgments (ACKs) indicate that a frame arrived correctly and in order; if a frame is damaged or out of order, the receiver discards it and all subsequent frames until the expected one arrives.
- In **Go-Back-N ARQ**, when a timer expires for a missing frame, the sender retransmits that frame and all following frames within the window.
- **Selective Repeat ARQ** resends only the damaged frames, using negative acknowledgments (NAKs) to request specific retransmissions, which is more bandwidth-efficient but requires more complex processing.
- **Piggybacking** combines data and acknowledgment in a single frame, saving bandwidth when both stations have data to send.

Flow control

Definition

Flow control is a set of procedures that tell the sender how much data it can transmit before it must wait for an acknowledgement from the receiver. It coordinates the amount of data that can be sent, whether measured in bits, bytes, kilobytes or frames, to match the receiver’s ability to process incoming data.

Explanation

The receiver has a limited processing speed and a finite amount of memory to store incoming data. When the receiver’s buffer approaches its limit, it must inform the sender to reduce or temporarily stop transmission; conversely, if the receiver’s buffer is under-utilized, it can request the sender to increase the rate. This feedback is typically conveyed through a advertised window size, which indicates the available buffer space at the receiver. The sender then limits its transmission rate to the minimum of the receiver’s advertised window and the network’s capacity, preventing buffer overflow and ensuring smooth data flow.

Example

TCP uses a sliding-window flow control algorithm combined with the go-back-N ARQ principle. The sender may transmit a number of frames equal to the current window size and then must wait for acknowledgements before sending additional frames.

Key points

- Flow control prevents the sender from overwhelming the receiver’s buffer by limiting the amount of unacknowledged data.
- The receiver advertises its available buffer space, and the sender adjusts its transmission rate accordingly.

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- In TCP, the sliding-window mechanism implements flow control together with error-recovery (go-back-N).

Error control

Definition

Error control in the data link layer consists of error detection and error correction mechanisms that allow the receiver to inform the sender about lost or damaged frames and to coordinate their retransmission.

Explanation

The process works by using acknowledgments (ACKs) and timers. The sender keeps a copy of the transmitted frame and starts a timer; if an ACK is not received before the timer expires, the sender assumes the frame was lost or corrupted and retransmits it. The receiver maintains a variable that indicates the sequence number of the next expected frame; when a correct frame arrives it sends a positive ACK containing the number of the next expected frame. This exchange forms an automatic repeat request (ARQ) scheme, which can be implemented with mechanisms such as Stop-and-Wait, Go-Back-N, and Selective Repeat.

Example

In the Stop-and-Wait ARQ protocol, the sender transmits a frame (sequence 0), waits for an ACK (sequence 1); upon receiving the ACK it sends the next frame (sequence 1) and expects an ACK (sequence 0), and so on. If the ACK is lost, the sender’s timer expires and the same frame is resent.

Algorithm / Steps

1. Sender transmits a frame and records its sequence number (0 or 1).
2. Sender starts a timer for the transmitted frame.
3. Receiver checks the incoming frame; if it is error-free, it sends a positive ACK with the number of the next expected frame.
4. If the sender receives the ACK before the timer expires, it updates its control variable to the next sequence number and sends the next frame.
5. If the timer expires before an ACK is received, the sender retransmits the same frame.

Key points

- Error control combines detection and correction, enabling retransmission of lost or damaged frames.
- ARQ mechanisms (Stop-and-Wait, Go-Back-N, Selective Repeat) implement error control at layer 2.
- Positive acknowledgments and timers are core to the Stop-and-Wait ARQ process.
- The receiver’s expected-frame variable and the sender’s control variable synchronize the exchange of frames and ACKs.

Data-Link Layer Protocols

Definition

The data-link layer, often paired with the physical layer, provides the predominant Ethernet protocol and includes two sub-layers: the Medium Access Control (MAC) sub-layer for media access mechanisms and the Logical Link Control (LLC) sub-layer that defines service access points to the upper layer.

Explanation

The MAC sub-layer handles the negotiation with the underlying physical medium (wired, wireless, etc.) and manages how multiple stations share the link, while the LLC sub-layer is concerned with communicating with higher layers such as the network layer and delivering link-level protocol data units (PDUs) between two stations without requiring intermediate switching. Network adapters (LAN cards) implement these data-link functions, interfacing with various physical media (e.g., Ethernet, Wi-Fi, Bluetooth) and enabling end-to-end connectivity through the LAN infrastructure.

Example

Ethernet is the major data-link protocol used in LANs, employing the MAC sub-layer for access control and the LLC sub-layer for higher-layer interaction.

Key points

- The data-link layer consists of MAC (media access) and LLC (service interface) sub-layers.
- MAC manages access to the shared physical medium; LLC handles communication with the network layer.
- LAN adapters provide the data-link and part of the physical layer functionality, supporting various media types.
- Data-link protocols enable transmission of link-level PDUs between two stations without intermediate switches.

PPP

Definition

PPP (Point-to-Point Protocol) is mentioned as a type of network that can be connected through a router using proxy ARP, allowing hosts on that network to communicate as if they were on the same physical LAN.

Explanation

When a router runs proxy ARP, it intercepts ARP requests from hosts on a PPP network and replies with its own MAC address. This makes the PPP hosts believe the destination is on the same local segment, even though the actual destination may reside on a different network. The router thus “proxies” ARP for both sides of the connection, simplifying address handling across the PPP link.

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Example

Consider two networks, A and B, linked by a router R that runs proxy ARP. Network B is a PPP network. A host on network B sends an ARP request for an IP address that actually belongs to network A. Router R replies with its MAC address, causing the host to place the router's MAC in its ARP table for that IP, and the communication proceeds through the router.

Key points

- Proxy ARP enables a router to answer ARP requests on behalf of hosts in a PPP network.
- The router's MAC address is stored in the ARP table of the PPP host for the destination IP.
- This mechanism makes hosts on different physical networks behave as if they share the same LAN, simplifying network configuration.

Media Access Control

Definition

Media Access Control (MAC) is the sub-layer of the data link layer that provides the mechanism for controlling access to the shared transmission medium, enabling devices to transmit frames efficiently over a LAN.

Explanation

The MAC sub-layer implements a controlling mechanism that determines **where** and **how** devices may place their frames onto the shared medium. It can operate in a centralized fashion, where a controller grants access, or in a distributed manner where each station follows a common protocol. The MAC is responsible for assembling data into frames that include destination addressing and error-detection fields, and for disassembling received frames while performing address recognition and error checking. By governing access to the medium, the MAC ensures that the available bandwidth is utilized maximally and that multiple devices can share the link without collisions.

Example

The predominant implementation of MAC in LANs is the Ethernet protocol, which provides the frame format, addressing scheme, and access methods used by most wired networks.

Key points

- MAC controls access to the shared transmission medium, enabling efficient use of bandwidth.
- It can be centralized (with a controller) or distributed (each station follows the protocol).
- Functions include frame assembly/disassembly, address recognition, and error detection.

- MAC is primarily used in broadcast and shared-channel networks such as Ethernet.

Ethernet Basics

Definition

Ethernet is the predominant data-link layer protocol defined by the IEEE 802 standards, specifically 802.3, and uses a CSMA/CD access method for wired LANs.

Explanation

Ethernet devices contain a MAC sub-layer that controls access to the shared transmission medium, ensuring efficient use of the available bandwidth. The MAC protocol decides where and how a device may transmit, either by a centralized controller or a distributed mechanism, to avoid conflicts on the shared media. Flow control is required because the receiver may process data slower than the sender transmits; the receiver signals the sender to adjust the sending rate or pause transmission to prevent buffer overflow.

Example

When a host needs to send an IP packet to a destination whose hardware address is unknown, it issues an ARP request. The ARP response provides the hardware address, after which the IP packet becomes the payload of an Ethernet frame and is sent as a unicast to the resolved destination. This illustrates how Ethernet frames encapsulate higher-layer packets and use hardware addresses for delivery.

Key points

- Ethernet is the main LAN protocol and follows the IEEE 802.3 (CSMA/CD) standard.
- The MAC sub-layer manages access to the shared medium, either centrally or in a distributed fashion, to maximize bandwidth usage.
- Flow control mechanisms allow the receiver to request the sender to reduce or stop transmission when its buffer limits are approached.
- ARP resolves IP addresses to Ethernet hardware addresses, enabling the formation of Ethernet frames for unicast delivery.

CSMA/CD

Definition

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the predominant Ethernet protocol defined in the IEEE 802.3 standard for the physical and data-link layers, providing a method for devices to share a common transmission medium while detecting collisions.

Explanation

In Ethernet networks, each station first listens to the carrier to determine if the medium is idle (carrier sense). If the channel is free, the station begins transmission. While transmitting, the station continues to monitor the medium; if it detects another transmission overlapping its own, a collision is recognized (collision detection). Upon detecting a collision, the stations stop transmitting and initiate a recovery procedure defined by the standard. This mechanism enables interoperability among devices from different manufacturers by adhering to the IEEE 802.3 specifications.

Example

Ethernet links operating at 10 Mbps, 100 Mbps, or Gigabit speeds all employ CSMA/CD as defined by IEEE 802.3, allowing devices such as computers and switches from various vendors to communicate over the same physical medium.

Key points

- CSMA/CD is the core access method for the Ethernet standard (IEEE 802.3).
- It combines carrier sensing before transmission with collision detection during transmission.
- The protocol ensures inter-operability of network devices from different manufacturers.
- Ethernet speeds ranging from 10 Mbps to several Gbps all rely on CSMA/CD.

Wireless LAN (802.11)

Definition

Wireless LAN (WLAN) is a separate consideration at the data link (layer 2) and physical (layer 1) layers, providing network connectivity without extensive cabling and enabling omnipresent networking for mobile devices.

Explanation

WLANs are based on IEEE 802.11 standards, which define various physical layer technologies (e.g., infrared, FHSS, DSSS, OFDM) and data rates. 802.11a operates in the 5 GHz ISM band with up to 54 Mbps, while 802.11b/g use the 2.4 GHz band with similar rates. 802.11n and later versions extend these capabilities. The architecture can be infrastructure-based, where all stations communicate through an access point (base station) that connects to the wired backbone. At the MAC layer, mechanisms such as RTS/CTS are used to mitigate hidden-node problems, though collisions may still occur in the noisy, mobile wireless environment.

Example

A typical WLAN deployment uses an 802.11a access point operating at 5 GHz, providing up to 54 Mbps data rate to laptops and smartphones within a range of about 20 meters.

Key points

- WLANs require network adapters (NICs) that handle both data link and physical layer functions for wireless media.
- IEEE 802.11 defines multiple physical layer options (infrared, FHSS, DSSS, OFDM) with varying frequencies, data rates, and range characteristics.
- Infrastructure mode uses an access point as a base station linking wireless stations to the wired network.
- RTS/CTS mechanisms are employed to address hidden-node issues, though collisions can still happen in mobile, noisy environments.

Physical Layer: Data and Signals

Definition

The physical layer is primarily responsible for the transmission of binary data through a medium, handling the actual communication phenomena that enable data transfer between nodes.

Explanation

It deals with encoding and decoding of data, generation of preambles, bit transmission and reception, and the characteristics of the transmission media and topology. These functions are governed by standards such as the IEEE 802 series. The layer also interfaces with network interface cards (NICs) to connect to various media types, including wired, wireless, Bluetooth, Zigbee, or satellite links.

Example

A LAN network adapter (NIC) in a PC enables the device to communicate over a shared Ethernet LAN by interfacing with the physical medium, whether it is a copper cable or a wireless link.

Key points

- Transmits binary data over wired or wireless media.
- Performs encoding/decoding, preamble generation, and bit-level transmission/reception.
- Interfaces with NICs to connect the computer to the physical medium (copper, wireless, satellite, etc.).
- Functions are defined by IEEE 802 standards covering both data link and physical layer aspects.

Performance

Definition

Performance refers to how efficiently a network delivers data, considering factors such as bandwidth utilization, round-trip time variability, and the impact of device choices on overall throughput and reliability.

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Explanation

Network performance is influenced by the underlying physical and data-link layers. At the physical layer, the total bandwidth of a shared medium is divided among all connected ports, so adding more devices can reduce the effective bandwidth per device and increase cost, affecting overall efficiency. At the data-link layer, the round-trip time (RTT) between two directly connected nodes shows low variance because only a single link is involved, allowing a simple average to estimate RTT accurately. In contrast, across multiple network layers, RTT variance is high due to dynamic intermediate paths, making static timeout settings unreliable and necessitating adaptive algorithms that continuously measure network performance. These aspects together determine the network's ability to support services such as cloud computing and high-performance applications, where any interruption can halt industrial processes, highlighting the critical need for robust performance management.

Example

When two nodes are connected through a single hub, the hub's total bandwidth is shared among all its ports. If the hub has four ports and a 100 Mbps link, each port effectively receives only 25 Mbps, illustrating how device choice directly impacts performance.

Key points

- Bandwidth of a shared medium is divided among all ports, reducing per-port throughput as more devices are added.
- RTT on a single link exhibits low variance, allowing accurate average-based estimation; multi-hop paths show high RTT variance, requiring dynamic timeout algorithms.
- Reliable performance is essential for services like cloud and high-performance computing; network interruptions can halt critical industrial processes.

Transmission media

Definition

The physical layer is responsible for the transmission of binary data through a medium, which may be wired, wireless, or any other form that allows communication between nodes.

Explanation

Transmission media can be used in two modes:

1. **Baseband transmission** – employs digital signalling over a single wire or channel, where the digital signal occupies the entire bandwidth of the medium and carries only one data signal. Typical multiplexing for baseband is Time Division Multiplexing (TDM).

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2. **Broadband transmission** – uses analog signals in the form of optical or electromagnetic waves over multiple transmission frequencies (carriers). Multiple signals can be carried simultaneously, requiring the media to be split into separate channels for sending and receiving.

Both modes rely on appropriate multiplexing techniques (e.g., a multiplexer combines multiple signals for broadband) and require the media to support the chosen signalling method.

Example

A baseband system transmits digital pulses over a single twisted-pair cable, using the whole bandwidth for one data stream.

Key points

- The physical layer handles data transmission through various media such as wired or wireless links.
- Baseband uses digital signals on a single channel, occupying the full bandwidth.
- Broadband uses analog signals on multiple frequency carriers, allowing simultaneous transmission of several signals.
- Multiplexers are used to combine multiple signals for broadband transmission.

Switching

Definition

Switching refers to the layer-2 mechanism that forwards frames from one part of a network to another, typically using a multi-port bridge or layer-2 switch. It operates at the data link layer to move frames toward their next hop while handling logical address resolution to physical addresses.

Explanation

A switch breaks up the collision domain by providing a separate collision domain for each of its ports, unlike a hub where all ports share a single collision domain. This isolation reduces collisions, retransmissions, and improves bandwidth utilization. Switches also perform frame forwarding based on MAC addresses, enabling efficient delivery of frames within the same LAN or toward other network segments.

Example

Consider a four-port hub operating at 100 Mbps. All four connected devices share one collision domain, leading to frequent collisions and an effective bandwidth of about 25 Mbps when all transmit simultaneously. Replacing the hub with a four-port layer-2 switch gives each port its own collision domain, eliminating those collisions and preserving the full 100 Mbps bandwidth for each link.

Key points

- Each port on a layer-2 switch constitutes its own collision domain, preventing collisions across ports.

- Switches forward frames based on MAC addresses, enabling layer-2 forwarding between devices.
- Replacing hubs with switches improves bandwidth efficiency and reduces retransmissions.
- Switches are multi-port bridges that can also connect to higher-level networks such as the Internet cloud.

Circuit Switching.

Definition

Circuit switching is a switching method that was developed alongside packet switching and is distinguished from it in networking discussions.

Explanation

In the early 1960s, networking research explored both packet switching, where data is divided into small packets that may travel different routes, and circuit switching as an alternative approach. The lecture notes indicate that circuit switching will be covered in subsequent sessions, highlighting its role as a counterpart to packet switching.

Key points

- Circuit switching is one of the two primary switching paradigms discussed in early networking history.
- It is presented as a contrast to packet switching, which splits data into small packets that can follow different paths.

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